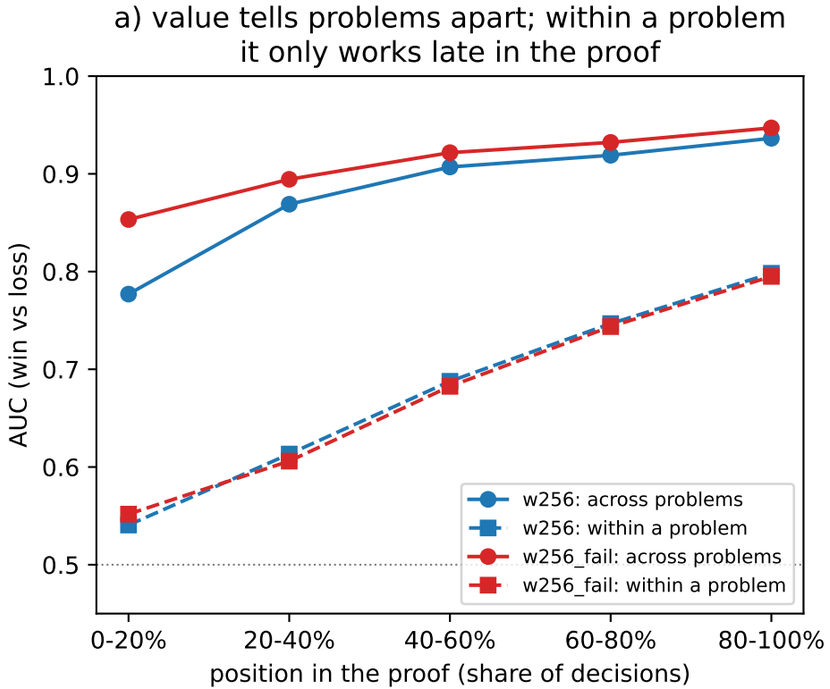
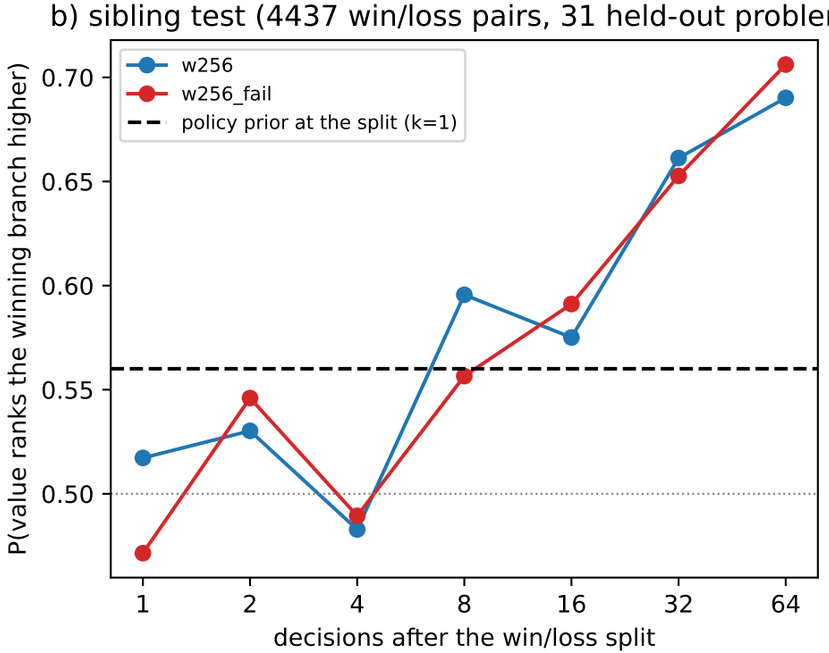
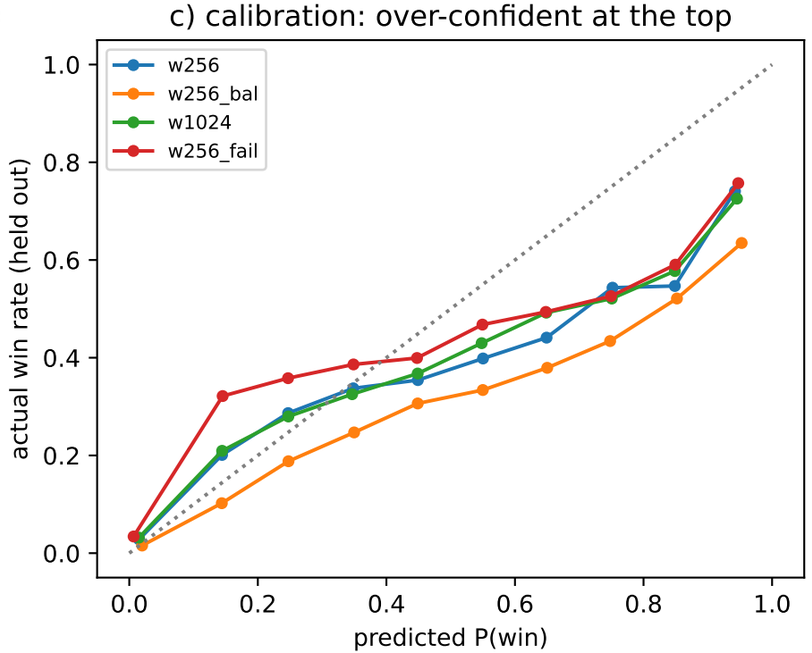

a) value tells problems apart; within a problem it only works late in the proof
AUC (win vs loss)
1.0
0.9
0.8
0.7
0.6
0.5
w256: across problems
w256: within a problem
w256_fail: across problems
w256_fail: within a problem
0-20%
20-40%
40-60%
60-80%
80-100%
position in the proof (share of decisions)


b) sibling test (4437 win/loss pairs, 31 held-out problems)
P(value ranks the winning branch higher)
0.70
0.65
0.60
0.55
0.50
w256
w256_fail
policy prior at the split (k=1)
1
2
4
8
16
32
64
decisions after the win/loss split


c) calibration: over-confident at the top
actual win rate (held out)
1.0
0.8
0.6
0.4
0.2
0.0
w256
w256_bal
w1024
w256_fail
0.0
0.2
0.4
0.6
0.8
1.0
predicted P(win)